#### 14.12.6 Board-Based Interpretation

These same relationships can be seen directly on an options board. As the future moves up through a strike region, lower-strike calls become more in the money first, while higher-strike calls lag behind. That difference is exactly what creates the value of a call spread.

Similarly, as the future moves down through a strike region, higher-strike puts gain intrinsic value before lower-strike puts do, which is what drives put spread value.

So the board is not just a list of separate options. The values across strikes are linked by hard payoff relationships.

#### 14.12.7 Other Important Structural Bounds

The same logic extends beyond simple vertical spreads.

**Butterflies.** A symmetric butterfly is built as

$$+1,\ -2,\ +1$$

across three equally spaced strikes. Its payoff can never be negative at expiration, so its fair value should never be negative before expiration either.

**Calendars.** A same-strike calendar spread compares two expirations on the same underlying. The longer-dated option should not be worth less than the shorter-dated one, since it gives the holder strictly more optionality.

**Put–call parity.** Calls, puts, strikes, and the underlying are connected by a strict relationship. If that relationship is violated, then some package of options and underlying can be bought and sold to lock in arbitrage.

These bounds are foundational because they let traders check prices quickly, estimate neighboring strikes, and recognize when a quoted market is clearly inconsistent.

## 15 Delta

### 15.1 Definition and Intuition

**Definition 15.1** (Delta)**. Delta**, denoted by $\Delta$, measures the sensitivity of an option's price to a change in the price of the underlying.

In practice, delta is the most immediate Greek for understanding directional exposure. It tells us how much the option price should change for a small move in the underlying, and it also tells us how much of the underlying we need to trade in order to hedge that option position.

**Remark 15.1.** Market makers usually think about delta first because it is the Greek most directly tied to short-term directional risk.

There are three common ways to interpret delta:

1. **Price sensitivity:** how much the option price changes for a small move in the underlying.
2. **Hedge ratio:** how many underlying contracts are needed to offset the option's directional exposure.
3. **Approximate probability of finishing in the money:** a rough rule of thumb often used by traders.

1) The delta of an option is the sensitivity of an option relative to changes in the price of an underlying asset. Delta describes how much the price of the option (or option strategy) will change as the underlying moves.
2) The hedge ratio of an option. The amount of an underlying asset we must trade to create a balanced portfolio that is indifferent to directional movements in the underlying.
3) The probability that an option will expire ITM. So, a 0.4 delta option would have about a 40% chance of finishing ITM.

Call Delta Calls Puts Put Delta

| Theta | Gamma | Vol | QBidO | cVolum | C Delta | Bid Qty | BidPx | Theo | Ask Px | Ask Qty | Strike | T Pos | Bid Qty | BidPx | Theo | Ask Px | Ask Qt | P Delta | pVolume | QBidO p |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| -0.002 | 0.0001 | 0.245 | 0.155 | | 1.00 | | | 109.02 | | | 260 | 0 | 0 | 0 | 0.02 | 0.125 | 25 | 0.00 | | 0.048 |
| -0.005 | 0.0003 | 0.245 | 0.155 | | 1.00 | | | 99.06 | | | 270 | 0 | 0 | 0 | 0.06 | 0.125 | 23 | 0.00 | | 0.048 |
| -0.010 | 0.0006 | 0.245 | 0.154 | | 0.99 | | | 89.16 | | | 280 | 0 | 25 | 0.13 | 0.16 | 0.250 | 22 | -0.01 | | 0.048 |
| -0.018 | 0.0011 | 0.245 | 0.154 | | 0.98 | | | 79.36 | | | 290 | 0 | 21 | 0.250 | 0.36 | 0.375 | 21 | -0.02 | | 0.049 |
| -0.027 | 0.0018 | 0.238 | 0.154 | | 0.97 | | | 69.64 | | | 300 | 0 | 20 | 0.625 | 0.64 | 0.750 | 20 | -0.04 | | 0.052 |
| -0.039 | 0.0027 | 0.231 | 0.152 | | 0.94 | | | 60.09 | | | 310 | 0 | 19 | 1.000 | 1.09 | 1.125 | 19 | -0.06 | | 0.055 |
| -0.053 | 0.0040 | 0.224 | 0.148 | | 0.91 | | | 50.82 | | | 320 | 0 | 18 | 1.750 | 1.82 | 1.875 | 18 | -0.09 | | 0.058 |
| -0.071 | 0.0055 | 0.220 | 0.145 | | 0.86 | | | 42.05 | | | 330 | 0 | 16 | 3.000 | 3.05 | 3.125 | 16 | -0.14 | | 0.062 |
| -0.089 | 0.0072 | 0.216 | 0.140 | | 0.79 | | | 33.87 | | | 340 | 0 | 15 | 4.750 | 4.87 | 4.875 | 15 | -0.21 | | 0.074 |
| -0.106 | 0.0088 | 0.213 | 0.135 | | 0.71 | | | 26.55 | | | 350 | 0 | 14 | 7.500 | 7.55 | 7.625 | 14 | -0.29 | | 0.091 |
| -0.118 | 0.0098 | 0.212 | 0.124 | | 0.61 | 11 | 20.250 | 20.27 | 20.375 | 11 | 360 | 0 | 13 | 11.250 | 11.27 | 11.375 | 13 | -0.39 | | 0.108 |
| -0.122 | 0.0103 | 0.211 | 0.112 | | 0.51 | 11 | 14.875 | 15.00 | 15.000 | 11 | 370 | 0 | 12 | 15.875 | 16.00 | 16.000 | 12 | -0.49 | | 0.112 |
| -0.120 | 0.0099 | 0.213 | 0.108 | | 0.41 | 12 | 10.875 | 10.94 | 11.000 | 12 | 380 | 0 | 11 | 21.875 | 21.94 | 22.000 | 11 | -0.59 | | 0.124 |
| -0.112 | 0.0091 | 0.215 | 0.098 | | 0.32 | 13 | 7.750 | 7.81 | 7.875 | 13 | 390 | 0 | | | 28.81 | | | -0.68 | | 0.135 |
| -0.099 | 0.0079 | 0.217 | 0.089 | | 0.25 | 14 | 5.375 | 5.49 | 5.500 | 14 | 400 | 0 | | | 36.49 | | | -0.76 | | 0.140 |
| -0.085 | 0.0065 | 0.221 | 0.080 | | 0.18 | 15 | 3.750 | 3.86 | 3.875 | 15 | 410 | 0 | | | 44.86 | | | -0.82 | | 0.145 |
| -0.071 | 0.0053 | 0.225 | 0.073 | | 0.14 | 16 | 2.625 | 2.70 | 2.750 | 16 | 420 | 0 | | | 53.70 | | | -0.86 | | 0.148 |
| -0.058 | 0.0042 | 0.229 | 0.067 | | 0.10 | 18 | 1.875 | 1.89 | 2.000 | 18 | 430 | 0 | | | 62.89 | | | -0.90 | | 0.150 |
| -0.047 | 0.0032 | 0.234 | 0.062 | | 0.07 | 19 | 1.250 | 1.34 | 1.375 | 19 | 440 | 0 | | | 72.34 | | | -0.93 | | 0.150 |
| -0.037 | 0.0025 | 0.238 | 0.058 | | 0.05 | 20 | 0.875 | 0.93 | 1.000 | 20 | 450 | 0 | | | 81.93 | | | -0.95 | | 0.150 |
| -0.028 | 0.0018 | 0.241 | 0.055 | | 0.04 | 21 | 0.625 | 0.63 | 0.750 | 21 | 460 | 0 | | | 91.63 | | | -0.96 | | 0.150 |
| -0.024 | 0.0014 | 0.248 | 0.053 | | 0.03 | 22 | 0.375 | 0.49 | 0.500 | 22 | 470 | 0 | | | 101.49 | | | -0.97 | | 0.150 |
| -0.021 | 0.0012 | 0.258 | 0.052 | | 0.02 | 23 | 0.375 | 0.40 | 0.500 | 23 | 480 | 0 | | | 111.40 | | | -0.98 | | 0.150 |
| -0.015 | 0.0008 | 0.258 | 0.050 | | 0.02 | 25 | 0.250 | 0.26 | 0.375 | 25 | 490 | 0 | | | 121.26 | | | -0.98 | | 0.150 |
| -0.010 | 0.0006 | 0.258 | 0.049 | | 0.01 | 25 | 0.125 | 0.16 | 0.250 | 25 | 500 | 0 | | | 131.16 | | | -0.99 | | 0.155 |
| -0.007 | 0.0004 | 0.258 | 0.049 | | 0.01 | 0 | 0.000 | 0.10 | 0.125 | 25 | 510 | 0 | | | 141.10 | | | -0.99 | | 0.155 |
| -0.005 | 0.0003 | 0.258 | 0.048 | | 0.01 | 0 | 0.000 | 0.06 | 0.125 | 25 | 520 | 0 | | | 151.06 | | | -1.00 | | 0.155 |
| -0.003 | 0.0002 | 0.258 | 0.048 | | 0.00 | 0 | 0.000 | 0.04 | 0.125 | 25 | 530 | 0 | | | 161.04 | | | -1.00 | | 0.155 |
| -0.002 | 0.0001 | 0.258 | 0.048 | | 0.00 | 0 | 0.000 | 0.02 | 0.125 | 25 | 540 | 0 | | | 171.02 | | | -1.00 | | 0.155 |
| -0.001 | 0.0001 | 0.258 | 0.048 | | 0.00 | 0 | 0.000 | 0.01 | 0.125 | 25 | 550 | 0 | | | 181.01 | | | -1.00 | | 0.155 |
| -0.001 | 0.0001 | 0.258 | 0.048 | | 0.00 | 0 | 0.000 | 0.01 | 0.125 | 25 | 560 | 0 | | | 191.01 | | | -1.00 | | 0.155 |

all options are assigned a delta between 0 and 1 Put options are assigned a delta betwee 0 and -1

**Figure 22:** Call deltas lie between 0 and 1, while put deltas lie between 0 and $-1$. On a screen, traders often use the delta columns to quickly gauge directional exposure across strikes.

### 15.2 Call and Put Delta Ranges

A call option always has positive delta:

$$0 \leq \Delta_{\text{call}} \leq 1.$$

A put option always has negative delta:

$$-1 \leq \Delta_{\text{put}} \leq 0.$$

Traders often drop the decimal and speak in whole-number terms. For example:

$$0.57 \text{ delta} \quad \text{is called} \quad 57 \text{ delta}.$$

Likewise, a put with delta $-0.20$ may simply be called a "20 delta put," with the negative sign understood from context.

**Remark 15.2.** Calls become more valuable as the underlying rises, so their deltas are positive. Puts become more valuable as the underlying falls, so their deltas are negative.

### 15.3 Delta as Price Sensitivity

The first and most literal interpretation of delta is that it approximates the change in option value for a small move in the underlying.

**Example 15.1** (Small move in the underlying)**.** Suppose the underlying is at \$100.00.

Let a call option be worth \$1.00 with delta $+0.30$, and let a put option be worth \$3.00 with delta $-0.70$.

If the underlying rises by \$0.20, then the new call value is approximately

$$1.00 + (0.20)(0.30) = 1.06,$$

while the new put value is approximately

$$3.00 + (0.20)(-0.70) = 2.86.$$

If instead the underlying falls by \$0.50, then the call value becomes approximately

$$1.00 + (-0.50)(0.30) = 0.85,$$

and the put value becomes approximately

$$3.00 + (-0.50)(-0.70) = 3.35.$$

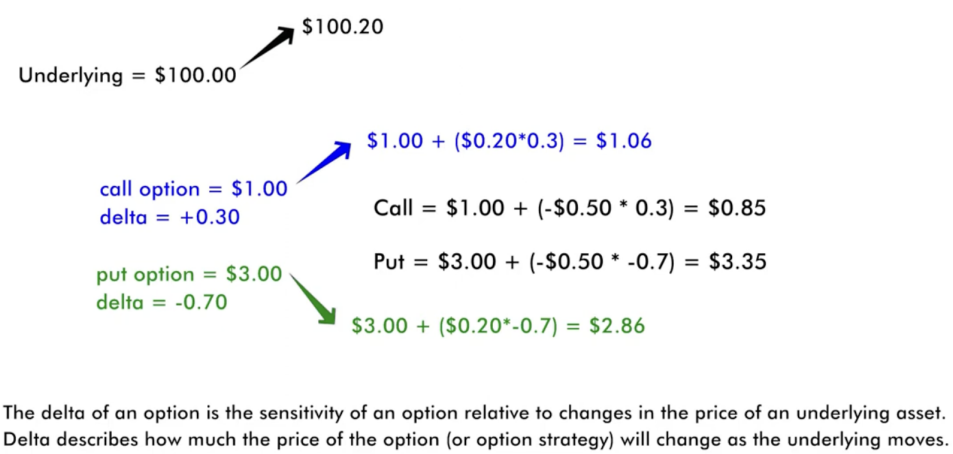


**Figure 23:** Example of delta as price sensitivity. A call with positive delta rises with the underlying, while a put with negative delta falls when the underlying rises and increases when the underlying falls.

**Remark 15.3.** This is only a local approximation. Delta works well for small underlying moves. For larger moves, the delta itself changes, which is exactly what gamma will later describe.

### 15.4 Delta as Hedge Ratio

The second interpretation of delta is as a hedge ratio. If an option position has directional exposure, we can trade the underlying against it to offset that exposure.

**Definition 15.2** (Delta hedge)**.** A **delta hedge** is a position in the underlying taken to offset the directional exposure of an option or option portfolio.

We can also think of the hedge rule as a sign calculation: the action on the option times the option's directional exposure gives the position's overall exposure, and the hedge must be taken in the opposite direction.

| Option Trade | (O Action | * | O Exposure) | = | Mult Result | | Hedge | |
|---|---|---|---|---|---|---|---|---|
| buy call | + | * | + | = | + | → | - | sell future |
| sell call | - | * | + | = | - | → | + | buy future |
| buy put | + | * | - | = | - | → | + | buy future |
| sell put | - | * | - | = | + | → | - | sell future |

**Figure 24:** The hedge can be understood as the opposite of the option position's directional exposure. This sign-based view makes the buy/sell hedge rule much easier to remember.

**Example 15.2** (Hedging a long put)**.** Suppose we buy 1 put with delta $-0.30$. Since we are long an option with negative directional exposure, our position is short delta overall. To hedge that, we buy 0.30 units of the underlying.

**Remark 15.4.** A delta-neutral hedge does not mean the position is riskless. It only means that for a small move in the underlying, the portfolio is approximately indifferent to direction.

### 15.5 Delta as Approximate ITM Probability

A third and more informal interpretation is that delta gives the approximate probability that an option finishes in the money.

So:

$$0.40 \text{ delta call} \approx 40\% \text{ chance of expiring ITM.}$$

Similarly:

$$-0.70 \text{ delta put} \approx 70\% \text{ chance of expiring ITM.}$$

**Remark 15.5.** This is a trader's rule of thumb, not a strict mathematical identity. It is intuitive and useful, but it becomes less exact once interest rates, dividends, and other pricing effects matter.

### 15.6 Call-Put Delta Relationship

For the same strike and expiration, call and put deltas are linked. Ignoring more advanced complications, we have the approximate relationship

$$\Delta_{\text{call}} + |\Delta_{\text{put}}| = 1.$$

**Example 15.3.** If a call has delta 0.91, then the corresponding put has delta approximately

$$-0.09.$$

**Remark 15.6.** This relationship is why many traders mainly watch one delta column and infer the other side mentally.

### 15.7 How Volatility Affects Delta

Delta is not fixed. It changes when model inputs change.

One important driver is implied volatility. When implied volatility rises, out-of-the-money option deltas tend to move closer to 0.50 in absolute terms because there is more chance that the option may finish in the money. When implied volatility falls, deltas move closer to their boundary values.

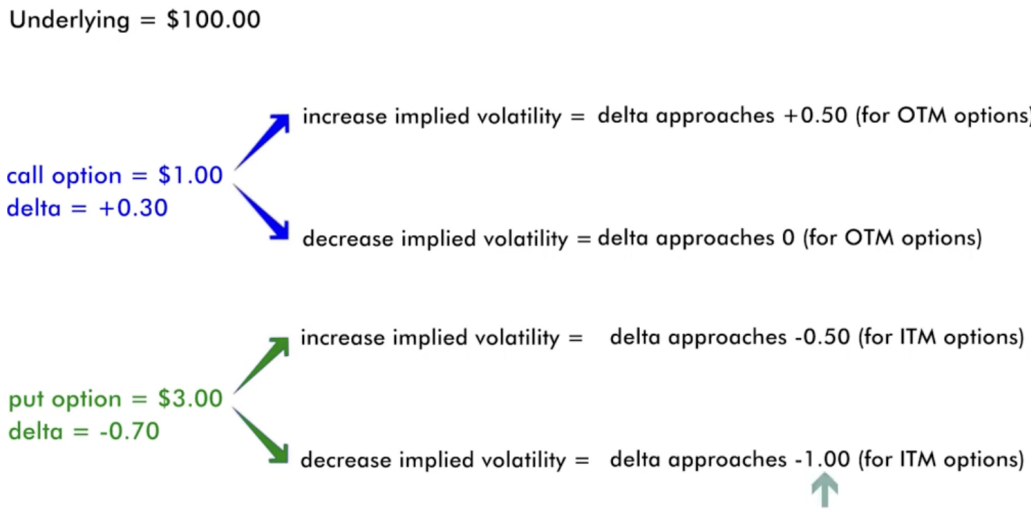


**Figure 25:** Effect of implied volatility on delta. Higher volatility pushes deltas toward the middle, while lower volatility pushes them toward 0 or $\pm 1$ depending on moneyness.

**Remark 15.7.** Higher volatility makes distant outcomes more plausible. That is why OTM options gain more "chance" and their deltas move inward toward 0.50 in absolute value.

### 15.8 How Time to Expiry Affects Delta

Time has a similar effect. More time to expiry gives the underlying more opportunity to move across strikes, so deltas tend to move toward the middle. Less time to expiry pushes deltas toward their expiration limits.

For example:

- an OTM call's delta moves toward 0.50 as time increases and toward 0 as expiry approaches,
- an ITM put's delta moves toward $-0.50$ as time increases and toward $-1$ as expiry approaches.

**Remark 15.8.** Increasing time and increasing implied volatility often affect delta in qualitatively similar ways: both make the option's final state less certain and therefore move delta toward the middle.

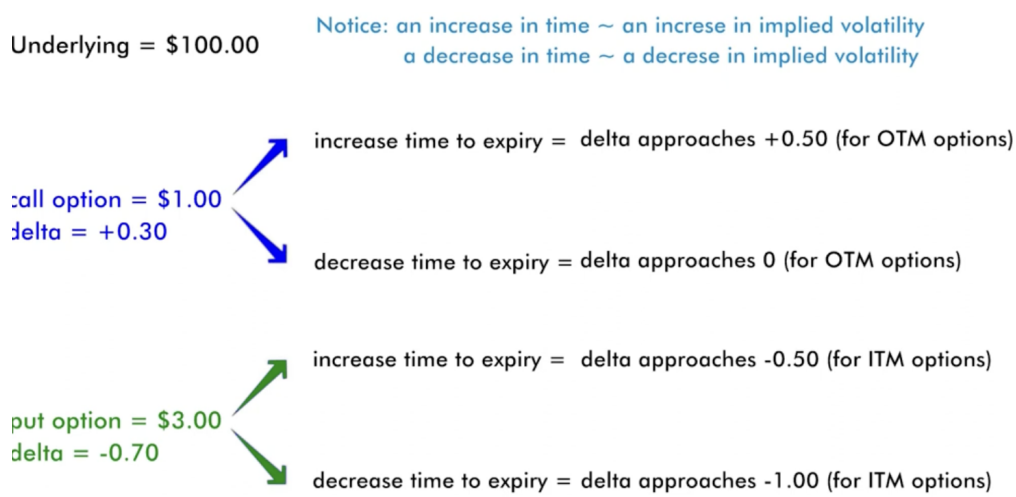


**Figure 26:** Effect of time to expiry on delta. More time pushes deltas toward the middle; less time pushes them toward expiration-style boundary behavior.

### 15.9 Covered or Tied Trades

There is also a practical trading application of delta called a **covered trade** or **tied trade**. In this structure, the option trade and the hedge trade are agreed as a package.

**Definition 15.3** (Covered trade)**.** A **covered trade** is a package in which an option is traded together with its corresponding hedge in the underlying.

**Remark 15.9.** This removes the immediate directional component from the trade because both sides receive the option and the hedge together.

**Figure 27:** Example of a tied or covered package: the buyer acquires calls and simultaneously sells futures, while the seller does the opposite.

**Example 15.4** (Covered call package)**.** Suppose a buyer purchases 50 calls at 2.75 and simultaneously sells 26 futures at 52.25 as the hedge. The seller takes the other side: short 50 calls and long 26 futures.

If multiple sellers are in the package market, the trade can be allocated across them according to their available quantities and priority.

**Remark 15.10.** Since futures cannot be traded fractionally, hedge quantities are rounded to whole futures in actual execution.